

Lecture 31/Module 3

Sinusoidal Steady State Analysis

22/09/2026

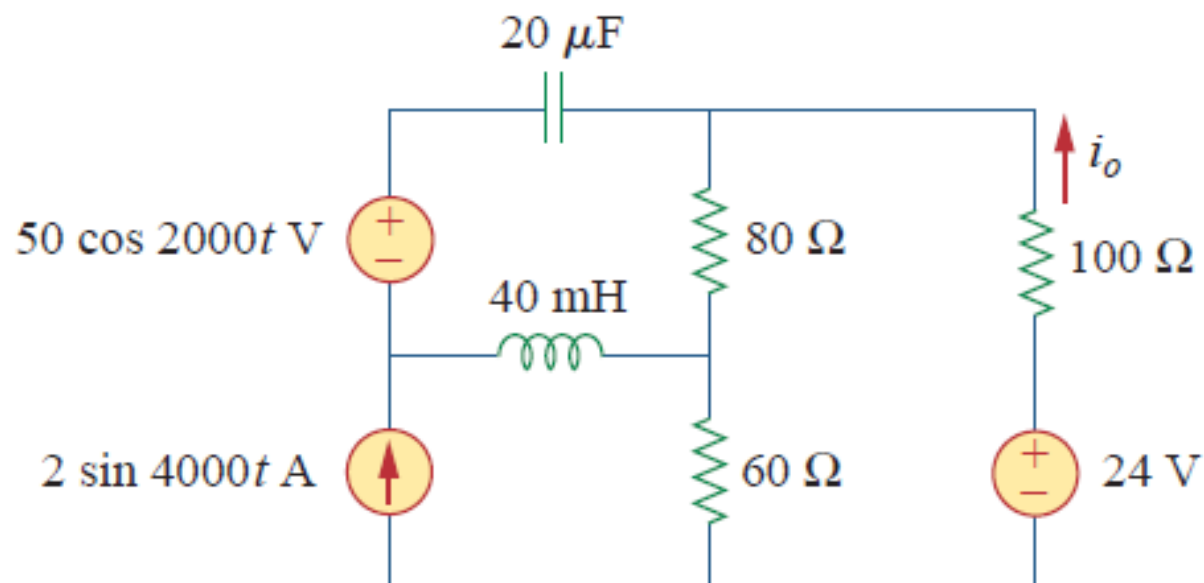
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10.48 Find i_o in the circuit of Fig. 10.93 using superposition.



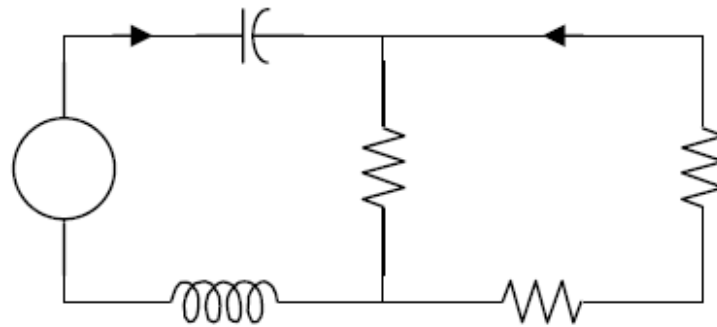
Let $i_o = i_{o1} + i_{o2} + i_{o3}$, where i_{o1} is due to the ac voltage source, i_{o2} is due to the dc voltage source, and i_{o3} is due to the ac current source. For i_{o1} , consider the circuit in Fig. (a).

$$\omega = 2000$$

$$50 \cos(2000t) \longrightarrow 50 \angle 0^\circ$$

$$40 \text{ mH} \longrightarrow j\omega L = j(2000)(40 \times 10^{-3}) = j80$$

$$20 \text{ } \mu\text{F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(2000)(20 \times 10^{-6})} = -j25$$



$$80 \parallel (60 + 100) = 160/3$$

$$\mathbf{I} = \frac{50}{160/3 + j80 - j25} = \frac{30}{32 + j33}$$

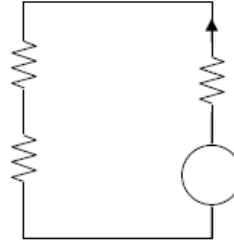
Using current division,

$$\mathbf{I}_{o1} = \frac{-80\mathbf{I}}{80+160} = \frac{-1}{3}\mathbf{I} = \frac{10\angle 180^\circ}{46\angle 45.9^\circ}$$

$$\mathbf{I}_{o1} = 0.217\angle 134.1^\circ$$

Hence, $i_{o1} = 0.217 \cos(2000t + 134.1^\circ) \text{ A}$

For i_{o2} , consider the circuit in Fig. (b).



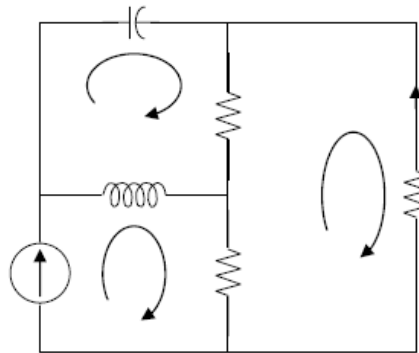
$$i_{o2} = \frac{24}{80+60+100} = 0.1 \text{ A}$$

For i_{o3} , consider the circuit in Fig. (c).

$$\omega = 4000$$

$$2 \cos(4000t) \longrightarrow 2\angle 0^\circ$$

$$40 \text{ mH} \longrightarrow j\omega L = j(4000)(40 \times 10^{-3}) = j160$$



$$20 \mu\text{F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(4000)(20 \times 10^{-6})} = -j12.5$$

$$\mathbf{I}_1 = 2 \quad (1)$$

For mesh 2,

$$(80 + j160 - j12.5)\mathbf{I}_2 - j160\mathbf{I}_1 - 80\mathbf{I}_3 = 0$$

Simplifying and substituting (1) into this equation yields

$$(8 + j14.75)\mathbf{I}_2 - 8\mathbf{I}_3 = j32 \quad (2)$$

For mesh 3,

$$240\mathbf{I}_3 - 60\mathbf{I}_1 - 80\mathbf{I}_2 = 0$$

Simplifying and substituting (1) into this equation yields

$$\mathbf{I}_2 = 3\mathbf{I}_3 - 1.5 \quad (3)$$

Substituting (3) into (2) yields

$$(16 + j44.25)\mathbf{I}_3 = 12 + j54.125$$

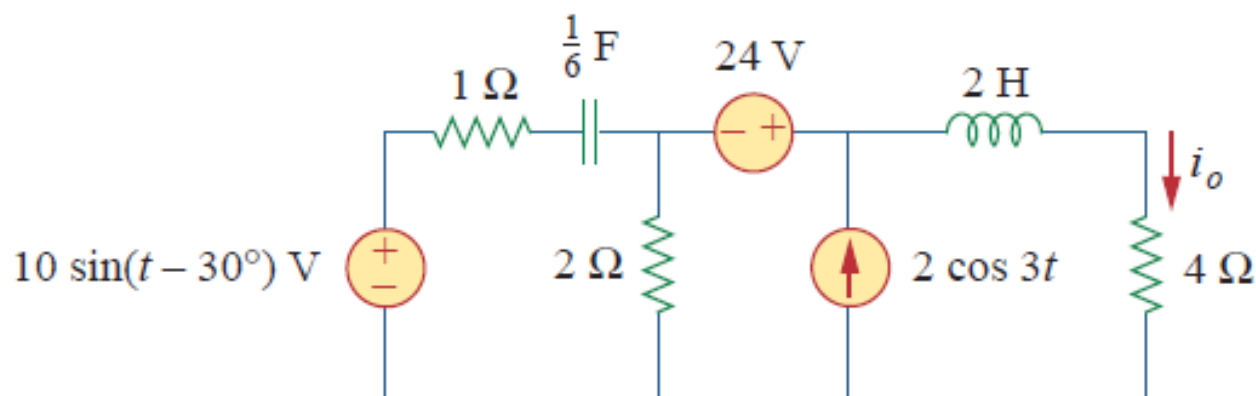
$$\mathbf{I}_3 = \frac{12 + j54.125}{16 + j44.25} = 1.1782 \angle 7.38^\circ$$

$$\mathbf{I}_{O3} = -\mathbf{I}_3 = -1.1782 \angle 7.38^\circ$$

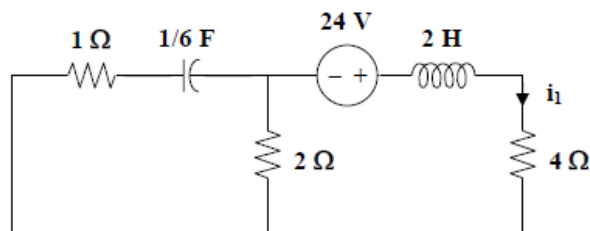
Hence, $i_{O3} = -1.1782 \sin(4000t + 7.38^\circ) \text{ A}$

Therefore, $i_O = \underline{0.1 + 0.217 \cos(2000t + 134.1^\circ) - 1.1782 \sin(4000t + 7.38^\circ) \text{ A}}$

10.47 Determine i_o in the circuit of Fig. 10.92, using the



Let $i_o = i_1 + i_2 + i_3$, where i_1 , i_2 , and i_3 are respectively due to the 24-V dc source, the ac voltage source, and the ac current source. For i_1 , consider the circuit in Fig. (a).

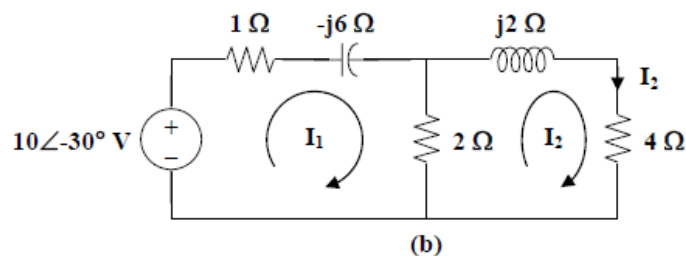


Since the capacitor is an open circuit to dc,

$$i_1 = \frac{24}{4+2} = 4 \text{ A}$$

For i_2 , consider the circuit in Fig. (b).

$$\begin{aligned} \omega &= 1 \\ 2 \text{ H} &\longrightarrow j\omega L = j2 \\ \frac{1}{6} \text{ F} &\longrightarrow \frac{1}{j\omega C} = -j6 \end{aligned}$$



For mesh 1,

$$\begin{aligned} -10\angle -30^\circ + (3 - j6)I_1 - 2I_2 &= 0 \\ 10\angle -30^\circ &= 3(1 - 2j)I_1 - 2I_2 \end{aligned} \quad (1)$$

For mesh 2,

$$\begin{aligned} 0 &= -2I_1 + (6 + j2)I_2 \\ I_1 &= (3 + j)I_2 \end{aligned} \quad (2)$$

Substituting (2) into (1)

$$10\angle -30^\circ = 13 - j15\mathbf{I}_2$$

$$\mathbf{I}_2 = 0.504\angle 19.1^\circ$$

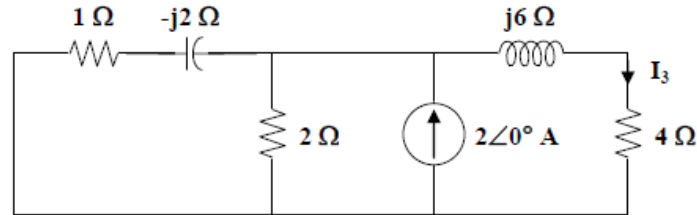
Hence, $i_2 = 0.504\sin(t + 19.1^\circ) \text{ A}$

For i_3 , consider the circuit in Fig. (c).

$$\omega = 3$$

$$2 \text{ H} \longrightarrow j\omega L = j6$$

$$\frac{1}{6} \text{ F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(3)(1/6)} = -j2$$



(c)

$$2 \parallel (1 - j2) = \frac{2(1 - j2)}{3 - j2}$$

Using current division,

$$\mathbf{I}_3 = \frac{\frac{2(1 - j2)}{3 - j2} \cdot (2\angle 0^\circ)}{4 + j6 + \frac{2(1 - j2)}{3 - j2}} = \frac{2(1 - j2)}{13 + j3}$$

$$\mathbf{I}_3 = 0.3352\angle -76.43^\circ$$

Hence $i_3 = 0.3352\cos(3t - 76.43^\circ) \text{ A}$

Therefore, $i_o = \underline{4 + 0.504\sin(t + 19.1^\circ) + 0.3352\cos(3t - 76.43^\circ) \text{ A}}$

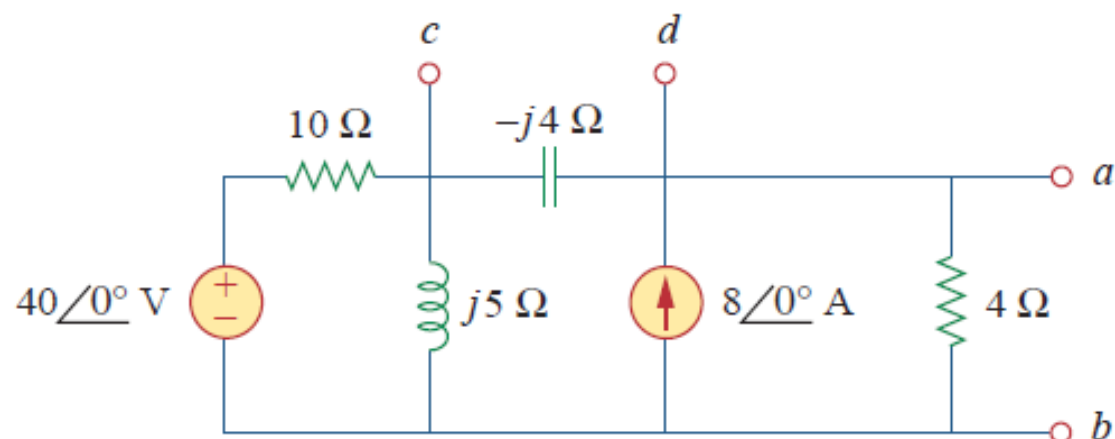
10.60 Find the Thevenin equivalent of the circuit in



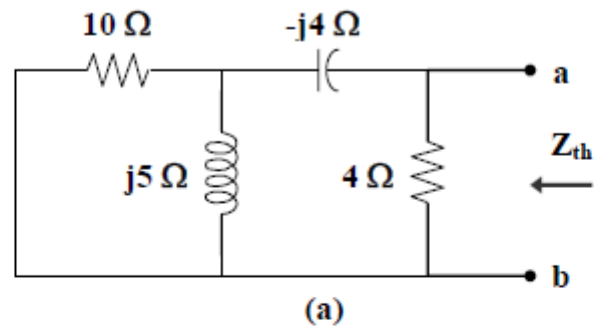
Fig. 10.103 as seen from:

(a) terminals a - b

(b) terminals c - d



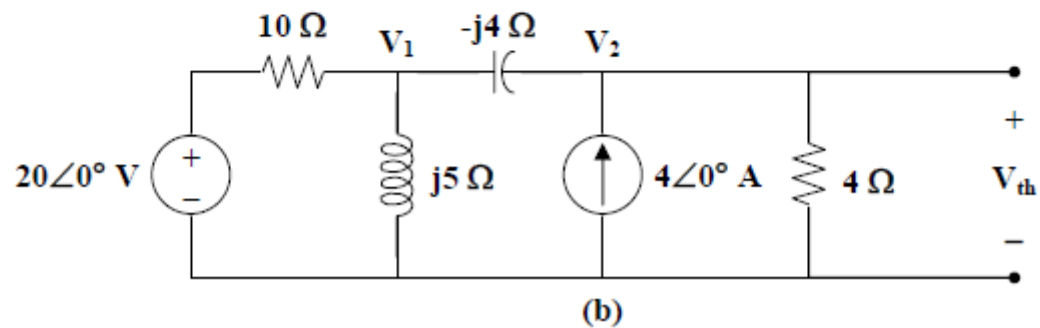
- (a) To find Z_{th} , consider the circuit in Fig. (a).



$$Z_{th} = 4 \parallel (-j4 + 10 \parallel j5) = 4 \parallel (-j4 + 2 + j4)$$

$$Z_{th} = 4 \parallel 2 = \underline{1.333\ \Omega}$$

- To find V_{th} , consider the circuit in Fig. (b).



At node 1,

$$\frac{20 - V_1}{10} = \frac{V_1}{j5} + \frac{V_1 - V_2}{-j4}$$

$$(1 + j0.5)V_1 - j2.5V_2 = 20$$
(1)

At node 2,

$$4 + \frac{V_1 - V_2}{-j4} = \frac{V_2}{4}$$

$$V_1 = (1 - j)V_2 + j16$$
(2)

Substituting (2) into (1) leads to

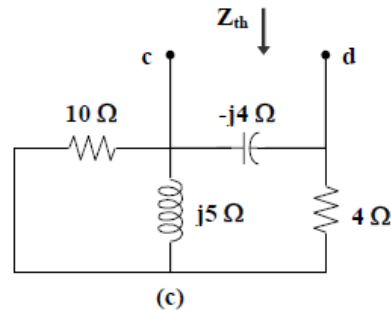
$$28 - j16 = (1.5 - j3)V_2$$

$$V_2 = \frac{28 - j16}{1.5 - j3} = 8 + j5.333$$

Therefore,

$$V_{th} = V_2 = \underline{9.615 \angle 33.69^\circ \text{ V}}$$

(b) To find Z_{th} , consider the circuit in Fig. (c).



$$Z_{th} = -j4 \parallel (4 + 10 \parallel j5) = -j4 \parallel \left(4 + \frac{j10}{2 + j}\right)$$

$$Z_{th} = -j4 \parallel (6 + j4) = \frac{-j4}{6} (6 + j4) = \underline{2.667 - j4 \Omega}$$

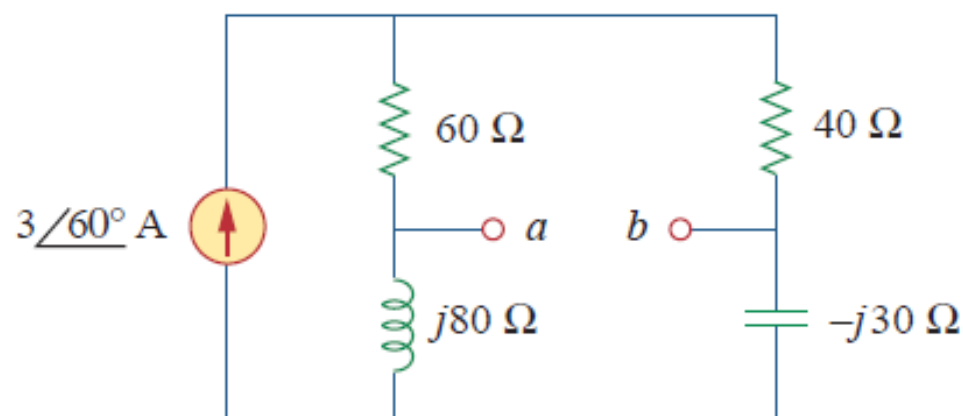
To find V_{th} , we will make use of the result in part (a).

$$V_2 = 8 + j5.333 = (8/3)(3 + j2)$$

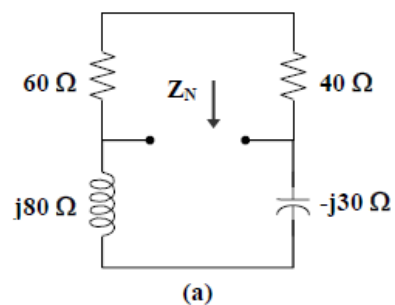
$$V_1 = (1 - j)V_2 + j16 = j16 + (8/3)(5 - j)$$

$$V_{th} = V_1 - V_2 = 16/3 + j8 = \underline{9.614 \angle 56.31^\circ \text{ V}}$$

10.64 For the circuit shown in Fig. 10.107, find the Norton equivalent circuit at terminals a - b .



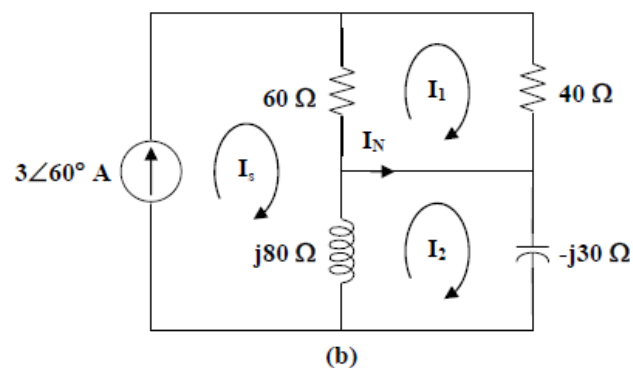
Z_N is obtained from the circuit in Fig. (a).



$$Z_N = (60 + 40) \parallel (j80 - j30) = 100 \parallel j50 = \frac{(100)(j50)}{100 + j50}$$

$$Z_N = 20 + j40 = \underline{44.72 \angle 63.43^\circ \Omega}$$

To find I_N , consider the circuit in Fig. (b).



$$I_s = 3 \angle 60^\circ$$

For mesh 1,

$$100I_1 - 60I_s = 0$$

$$I_1 = 1.8 \angle 60^\circ$$

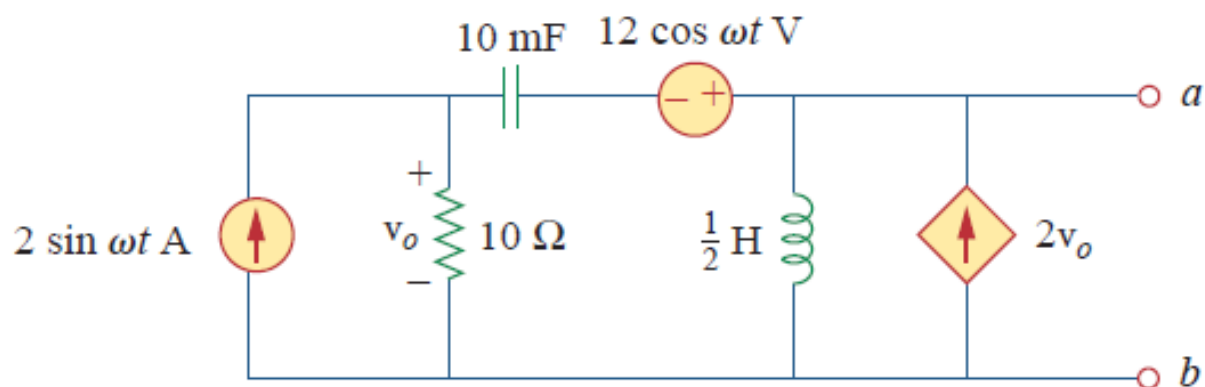
For mesh 2,

$$(j80 - j30)I_2 - j80I_s = 0$$

$$I_2 = 4.8 \angle 60^\circ$$

$$I_N = I_2 - I_1 = \underline{3 \angle 60^\circ \text{ A}}$$

- 10.66** At terminals a - b , obtain Thevenin and Norton equivalent circuits for the network depicted in Fig. 10.109. Take $\omega = 10$ rad/s.

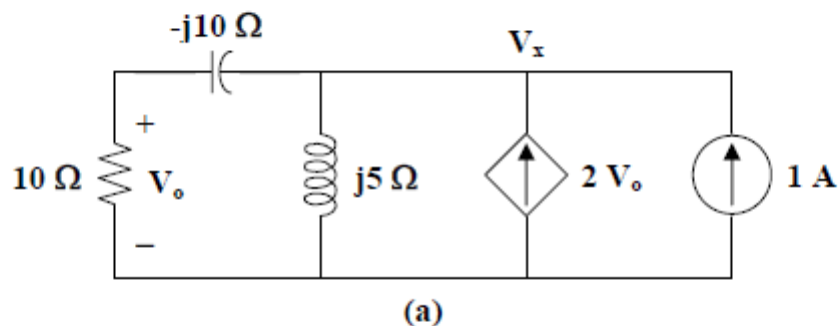


$$\omega = 10$$

$$0.5 \text{ H} \longrightarrow j\omega L = j(10)(0.5) = j5$$

$$10 \text{ mF} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(10)(10 \times 10^{-3})} = -j10$$

To find Z_{th} , consider the circuit in Fig. (a).



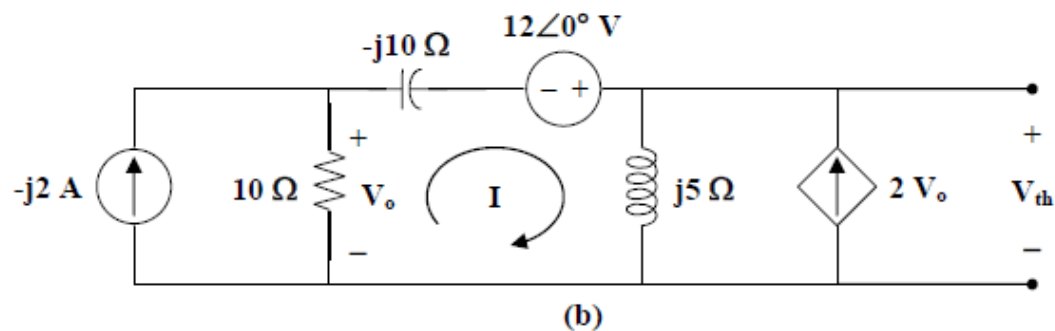
$$1 + 2V_o = \frac{V_x}{j5} + \frac{V_x}{10 - j10},$$

$$\text{where } V_o = \frac{10V_x}{10 - j10}$$

$$1 + \frac{19V_x}{10 - j10} = \frac{V_x}{j5} \longrightarrow V_x = \frac{-10 + j10}{21 + j2}$$

$$Z_N = Z_{th} = \frac{V_x}{1} = \frac{14.142 \angle 135^\circ}{21.095 \angle 5.44^\circ} = \underline{0.67 \angle 129.56^\circ \Omega}$$

To find V_{th} and I_N , consider the circuit in Fig. (b).



$$(10 - j10 + j5)I - (10)(-j2) + j5(2V_o) - 12 = 0$$

where

$$V_o = (10)(-j2 - I)$$

Thus,

$$(10 - j105)I = -188 - j20$$

$$I = \frac{188 + j20}{-10 + j105}$$

$$V_{th} = j5(I + 2V_o) = j5(-19I - j40) = -j95I + 200$$

$$V_{th} = \frac{-j95(188 + j20)}{-10 + j105} + 200 = 29.73 + j1.8723$$

$$V_{th} = \underline{29.79\angle 3.6^\circ \text{ V}}$$

$$I_N = \frac{V_{th}}{Z_{th}} = \frac{29.79\angle 3.6^\circ}{0.67\angle 129.56^\circ} = \underline{44.46\angle -125.96^\circ \text{ A}}$$

Thank You